

Challenge (Habanero)

- Q1.** A hiker is walking on a trail where the elevation y is given by $y = x^2 - 4x + 5$. If a stream runs along the trail with an elevation given by $y = 2x - 1$, at what point do they intersect?
- (A) (1, 1)
(B) (3, 5)
(C) (2, 1)
(D) (1, 3)
(E) (4, 5)
- Q2.** The population of rabbits P in a forest can be modeled by $P = 2x^2 + 5x + 1$, where x is the number of months since the initial count. If the population of foxes F can be modeled by $F = x + 2$, how many rabbits are there when there are 7 foxes?
- (A) 35
(B) 41
(C) 45
(D) 49
(E) 53
- Q3.** The fuel consumption F of a car is given by $F = x^2 - 2x + 1$, where x is the speed in miles per hour. If the distance traveled is given by $D = 2x + 1$, how far has the car traveled when it has consumed 5 gallons of fuel?
- (A) 5 miles
(B) 6 miles
(C) 7 miles
(D) 8 miles
(E) 9 miles
- Q4.** The cost C of a guided hike is given by $C = x^2 + 3x + 2$, where x is the number of hikers. If the revenue R is given by $R = 2x + 5$, how many hikers are needed to break even?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q5.** A park ranger is tracking the population of deer D given by $D = x^2 - x + 1$, where x is the number of months. If the population of wolves W is given by $W = x + 1$, what is the ratio of deer to wolves after 4 months?
- (A) 2:1
(B) 3:1
(C) 4:1
(D) 5:1
(E) 6:1
- Q6.** A group of friends want to go on a hot air balloon ride that costs $y = x^2 + 2x + 1$ dollars, where x is the number of people. If each person has a budget of $B = 2x - 1$, how many people can go on the ride?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** The temperature T in a cave is given by $T = x^2 - 3x + 2$, where x is the depth in feet. If the humidity H is given by $H = x + 1$, at what depth is the temperature 2 degrees higher than the humidity?
- (A) 2 feet
(B) 3 feet
(C) 4 feet
(D) 5 feet
(E) 6 feet
- Q8.** A farmer is planting crops in a field with an area given by $A = x^2 + 2x + 1$, where x is the length of the field. If the amount of fertilizer F needed is given by $F = x + 1$, how much fertilizer is needed for a field of length 3?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations: $y = x^2 - 2x + 1$ and $y = 3x - 2$. Find the intersection points and interpret the results in the context of a hiking trail where x is the distance traveled and y is the elevation.
- Q10.** A wildlife conservationist is tracking the population of a species of bird B given by $B = 2x^2 + x + 1$, where x is the number of years. If the population of a species of snake S is given by $S = x + 2$, find the ratio of birds to snakes after 5 years and interpret the results.

Chef's Tips

- Provide explicit instructions for each question.
- Ensure students have access to necessary materials and equipment.
- Encourage students to show their work and explain their reasoning.